

CS/SE/IT 1209

Operating Systems

Topic 1 : Introduction to Operating Systems

by K H Navoda [B.Sc. (Hons), MCS, BCS HEQ]

Bachelor of Science (Hons) in CS / SE / IT
Department of Computing
Faculty of Computing & Technology
Saegis Campus
Nugegoda.



What is an Operating System?

An OS is a fundamental software component that **manages computer hardware and software resources**, providing common services for computer programs.

Its primary purpose is to ensure the system operates **correctly and efficiently in an easy-to-use manner**.

There are two main views of an Operating System.

1. As an Extended Machine (Top-Down View)
2. As a Resource Manager (Bottom-Up View)

As an Extended Machine (Top-Down View)

The OS provides an **abstract layer** on top of bare hardware.

Hides hardware complexities from users and programmers

Offers a **convenient and user-friendly interface** via standard system calls to access devices

As a Resource Manager (Bottom-Up View)

Responsible for **managing and allocating system resources** (CPU time, memory, files, I/O devices)

Ensures **efficient and fair use** of these resources among competing processes and users

Purpose and Objectives of an OS

Convenience - Makes the computer system **easier and more user-friendly**

Efficiency - Ensures **efficient utilization of computer resources**

Ability to Evolve - Should be **modular** to allow for **new functions and updates** without disrupting service, adapting to changing hardware and user needs.

Protection - Safeguards information and resources of each process from unintended interference

Security - Guards against malicious applications and external threats

Reliability - Ensures data safety and consistency despite failures

Components of a Computer

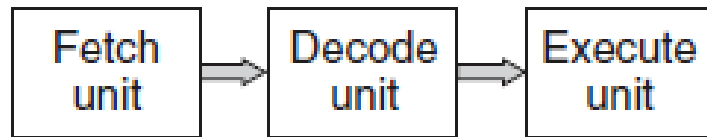
Processors

- The “brain” of the computer is the CPU. It fetches instructions from memory and executes them.
- The basic cycle of every CPU is to fetch the first instruction from memory, decode it to determine its type and operands, execute it, and then fetch, decode, and execute subsequent instructions.
- all CPUs contain some registers inside to hold key variables and temporary results.
- the instruction set generally contains instructions to load a word from memory into a register and store a word from a register into memory.
- Other instructions combine two operands from registers, memory, or both into a result, such as adding two words and storing the result in a register or in memory

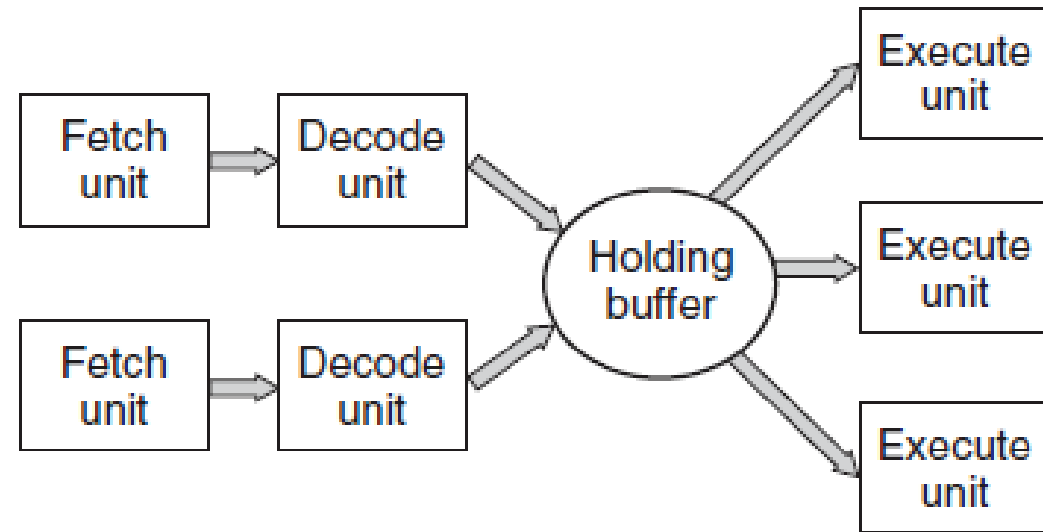
Processors

- To **improve performance**, CPU designers have long abandoned the simple model of fetching, decoding, and executing one instruction at a time.
- Many modern CPUs have facilities for executing more than one instruction at the same time.
- For example, a CPU might have separate fetch, decode, and execute units, so that while it is executing instruction n , it could also be decoding instruction $n + 1$ and fetching instruction $n + 2$.
- Such an organization is called a **pipeline**

Processors



(a)



(b)

Processors

- Even more advanced than a pipeline design is a superscalar CPU.
- In this design, multiple execution units are present
- For example, one for integer arithmetic, one for floating-point arithmetic, and one for Boolean operations.
- Two or more instructions are fetched at once, decoded, and dumped into a holding buffer until they can be executed.
- As soon as an execution unit becomes available, it looks in the holding buffer to see if there is an instruction it can handle, and if so, it removes the instruction from the buffer and executes it.

Multi Threaded and Multi Core Chips

- A thread is a kind of lightweight process
- If one of the processes needs to read a word from memory a multithreaded CPU can just switch to another thread.
- Multithreading does not offer true parallelism.
- Only one process at a time is running, but thread-switching time is reduced to the order of a nanosecond.
- Multithreading has implications for the operating system because each thread appears to the operating system as a separate CPU.
- Consider a system with two actual CPUs, each with two threads.
- The operating system will see this as four CPUs.

Multi Threaded and Multi Core Chips

- Beyond multithreading, many CPU chips now have four, eight, or more complete processors or cores on them.
- The multicore chips effectively carry four minichips on them, each with its own independent CPU.

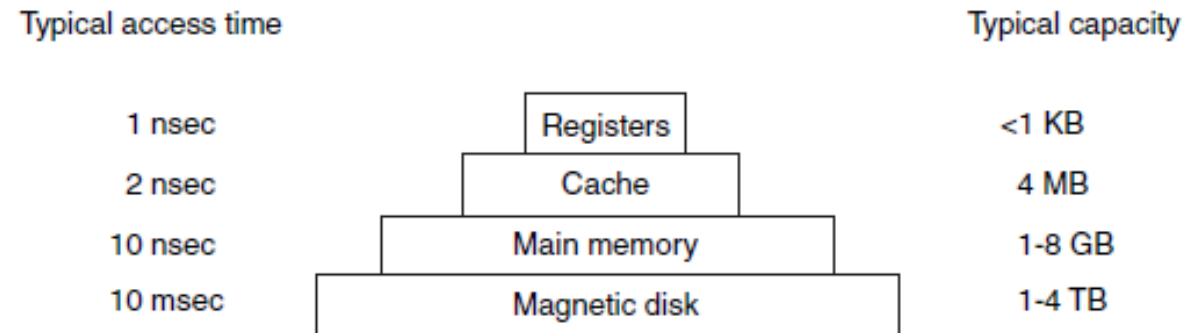
Memory

The second major component in any computer is the memory. Ideally, a memory

- Extremely fast
- Abundantly large, and
- Dirt cheap.

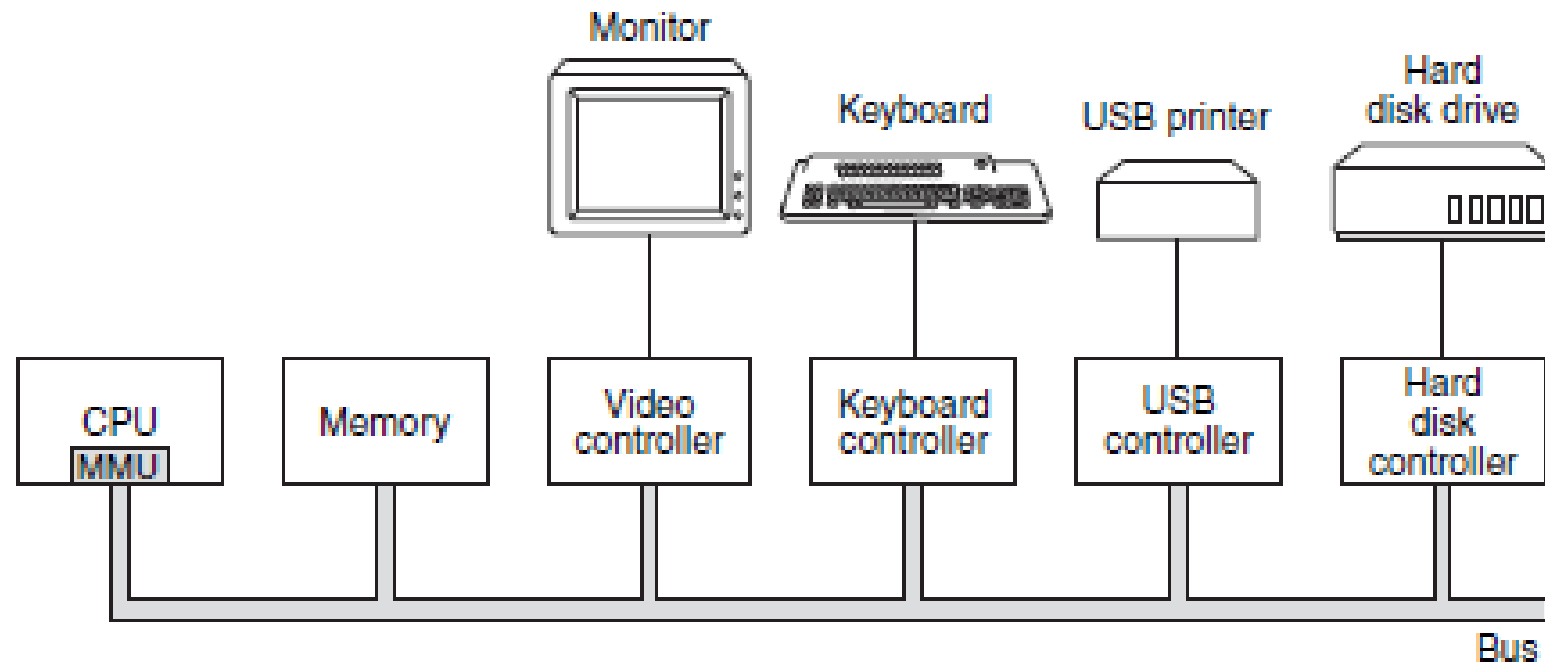
No current technology satisfies all of these goals, so a different approach is taken.

The memory system is constructed as a hierarchy of layers.



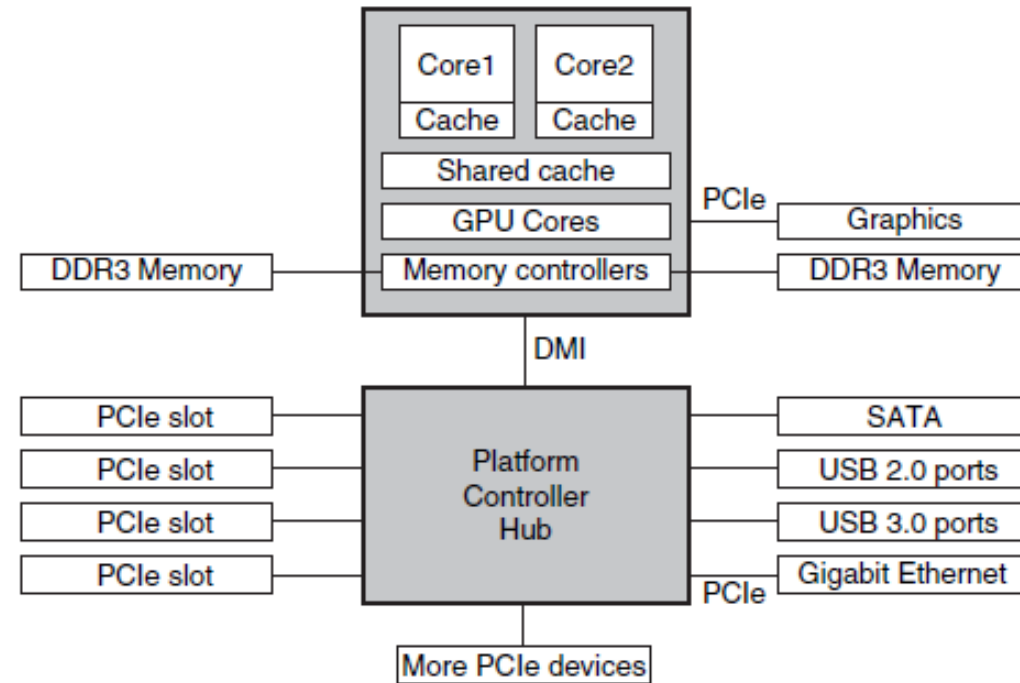
I/O Devices

- The CPU and memory are not the only resources that the operating system must manage. I/O devices also interact heavily with the operating system.



Buses

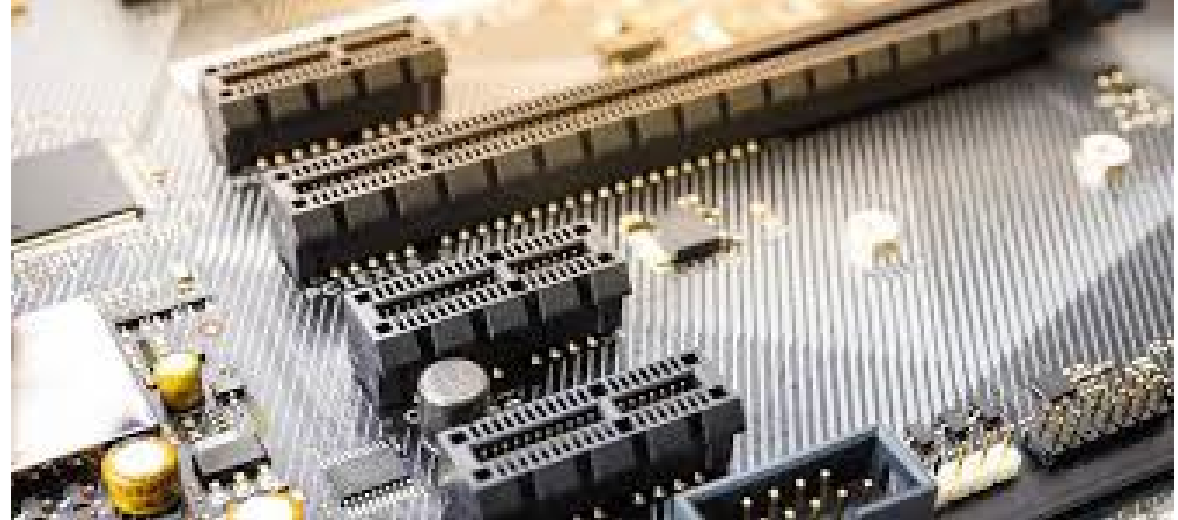
- A bus, in computing and digital technology, is an electronic pathway through which data can be transferred.
- This pathway uses signals that move at different speeds and are sent through different channels to communicate information between components within a computer or network.
- There are many system buses such as PCIe, PCI, USB, SATA and DMI.



Buses

PCIe (Peripheral Component Interconnect Express)

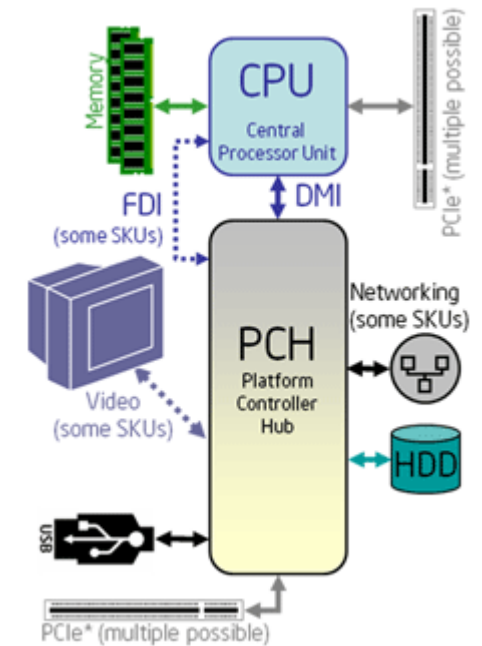
- Invented by Intel as a successor of PCI.
- Capable of transferring tens of gigabits per second.
- Uses a serial bus architecture



Buses

DMI (Direct Media Interface)

- A proprietary interconnect technology developed by Intel.
- It is primarily used to connect the CPU (Central Processing Unit) of a computer to the chipset, facilitating high-speed communication between the processor and other system components such as memory, peripherals, and graphics cards.



Buses

USB (Universal Serial Bus)

- Interface widely used standard for connecting peripherals to computers, tablets, and other electronic devices.
- It serves as a versatile and convenient method for transferring data and providing power to devices.
- Initially invented to attach all the slow I/O devices, such as the keyboard and mouse, to the computer.

Buses

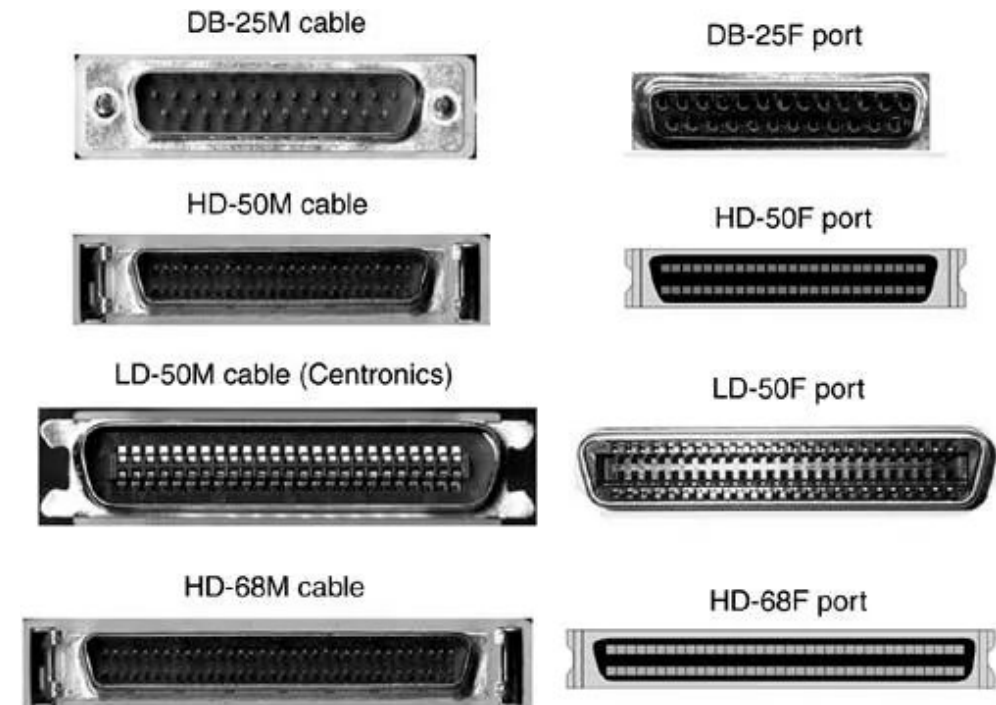
USB (Universal Serial Bus)



Buses

SCSI (Small Computer System Interface)

- Bus is a high-performance bus intended for fast disks, scanners, and other devices needing considerable bandwidth.
- Nowadays, find them mostly in servers and workstations.
- They can run at up to 640 MB/sec.



Operating Systems Categorization

Operating Systems Categorization

Operating systems are categorized in various ways, reflecting their design goals, architecture, and the computing environments they serve.

- Design Goal and Application Environment
 - Batch Systems
 - Time-Sharing Systems / Interactive Systems / Multiprogramming Systems / Multitasking Systems
 - Real-Time Operating Systems (RTOS)
 - General-Purpose Operating Systems
- Architecture and Structure
 - Monolithic Systems
 - Layered Systems
 - Microkernel Architectures
 - Client-Server Model
 - Hybrid Systems
 - Virtual Machines (VMs) / Virtualization
 - Exokernels

Operating Systems Categorization

- Number of Processors, Users, and Connectivity
 - Single-Processor Systems
 - Multiprocessor Systems / Parallel Computers / Multicomputer Systems
 - Single-User Systems
 - Multiuser Systems
 - Distributed Systems
- By Device Type and Scale
 - Mainframe Operating Systems
 - Server Operating Systems
 - Personal Computer (PC) Operating Systems
 - Handheld Computer Operating Systems / Mobile Computing

Operating System Structure

Monolithic Kernels

- The entire OS resides in the kernel (e.g., early Linux)
- All kernel code and data structures are in a single address space, avoiding context switches for OS functions

Microkernels

- Only core functions in the primary module; other modules run as user processes.
- More flexible but potentially less efficient due to message passing overhead

Hybrid Systems

- Combine aspects of both, often with a core kernel and **loadable kernel modules**.

Layered Design

- OS structured into levels; each level uses features of lower levels

Operating System Landscape

Mainframe Operating Systems

Designed for **room-sized computers** found in corporate data centers, characterized by prodigious I/O capacity and the ability to process many jobs simultaneously.

They typically offer **batch, transaction processing, and timesharing services**.

Examples include OS/390.



Server Operating Systems

Run on servers (large personal computers, workstations, or mainframes) to **serve multiple users over a network** and facilitate the sharing of hardware and software resources.

They can provide print, file, or Web services.

Examples include Solaris, FreeBSD, Linux, and Windows Server.



Multiprocessor Operating Systems

Designed for **systems with multiple CPUs or cores** (multicore systems) to achieve significant computing power.

These systems require specialized OS features for communication, connectivity, and consistency.

Symmetric Multiprocessing (SMP) is a common design where all processors are peers and share memory and devices.

Clustered systems are a specialized form where multiple networked computers cooperate to provide services, offering advantages in high availability and fault tolerance.

Personal Computer (PC) Operating Systems

Provide **strong support for a single user** and are widely used for applications like word processing, spreadsheets, games, and Internet access.

Modern PC OSES support multiprogramming, often running dozens of programs from boot time.

Examples include Linux, FreeBSD, Windows 7, Windows 8, and Apple's OS X.



Handheld Computer Operating Systems / Mobile Computing

Tailored for **tablets, smartphones, and other handheld devices**, featuring multicore CPUs, GPS, cameras, and sophisticated operating systems.

The market is dominated by **Google's Android** and **Apple's iOS**.



Embedded Operating Systems

Control devices not typically considered general-purpose computers, such as **microwave ovens, cars, or MP3 players**.

Software is often in ROM, and users typically cannot install new applications.

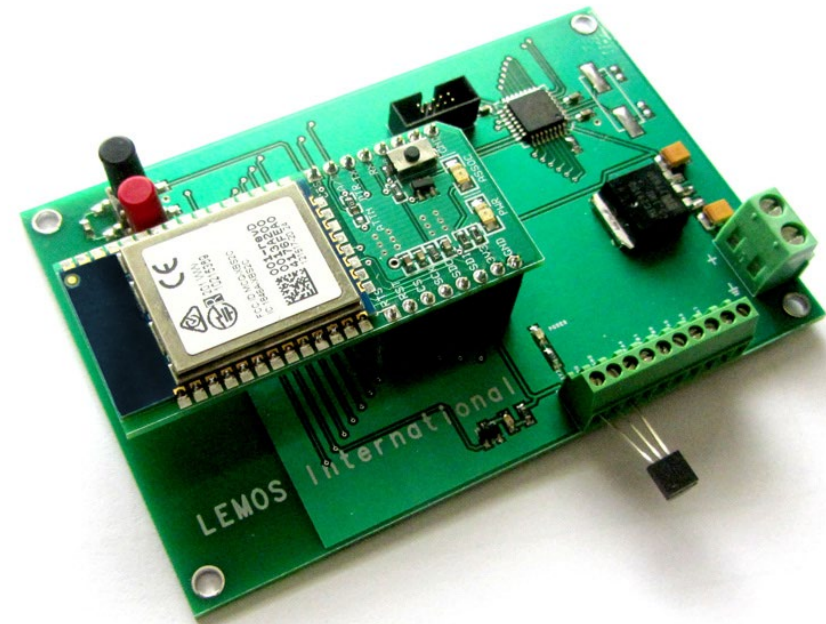
These systems often prioritize fast, lightweight process/thread switching, real-time scheduling, small size, and quick response to external interrupts.

Examples include Embedded Linux, QNX, VxWorks, eCos, and TinyOS.

Sensor-Node Operating Systems

For **small, real computers with CPU, RAM, ROM, and environmental sensors**, often event-driven, with limited memory and battery life.

TinyOS is a well-known example for wireless sensor networks.



Real-Time Operating Systems (RTOS)

Designed for **embedded environments where the correctness of a computation depends on the time it is delivered.**

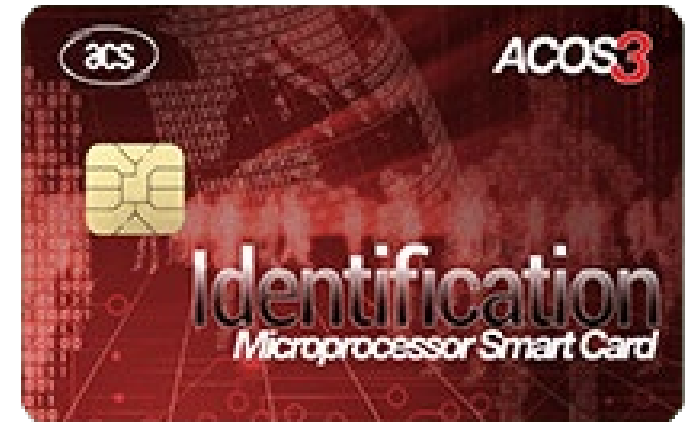
They are often event-driven and minimize intervals where interrupts are disabled to ensure quick responses, typically under 10 microseconds.

Examples include VxWorks, QNX, eCos, and TinyOS.

Smart Card Operating Systems

The **smallest type**, running on credit-card-sized devices with very severe processing power and memory constraints.

They may handle single or multiple functions, such as electronic payments.



Questions



